Week 13: Creativity and Machines

Philosophy of Artificial Intelligence

Philosophy of Artificial Intelligence (11118BLG003). Prof. Dr. Utku Kose. Full Professor, Department of Computer Engineering, Süleyman Demirel University, Isparta, Türkiye. Additional affiliations: University of North Dakota (USA), Universidad Panamericana (Mexico City, Mexico), Vel Tech University (Chennai, India). ORCID 0000-0002-9652-6415. E-mail: utkukose@sdu.edu.tr, utku.kose@und.edu, ukose@up.edu.mx, utkukose@gmail.com. This course is updated in line with current developments in the field. First version: September 2026. Last update: October 2026.

Overview

Can a machine originate anything, or can it only do what it is told? This week examines Lovelace's objection and Turing's reply, Boden's analysis of creativity, criteria for attributing creativity to programs and search methods that reward novelty [1, 2, 3].

Learning outcomes

By the end of the week, students are expected to state Lovelace's objection and Turing's reply. They should be able to distinguish combinational, exploratory and transformational creativity, and P-creativity from H-creativity. They should also be able to apply the criteria of novelty, quality and typicality. Finally, they should be able to explain how novelty search explores a conceptual space.

Lecture

Lovelace's objection

In her notes on Babbage's Analytical Engine, Ada Lovelace wrote that the engine had no pretensions to originate anything and could do only whatever people knew how to order it to perform [1]. Turing called this Lady Lovelace's objection and replied in two ways [4]. Machines often take people by surprise, because the consequences of their programs are not obvious to those who wrote them. More fundamentally, learning machines acquire behaviour that their programmers did not specify. The question for this week is whether surprise and learning amount to creativity.

Three kinds of creativity

Boden analysed creativity as the production of ideas that are new, surprising and valuable, and distinguished three ways in which such ideas arise [2, 5]. Combinational creativity produces unfamiliar combinations of familiar ideas, as in poetic imagery or analogy. Exploratory creativity explores a structured conceptual space, such as a musical style, and finds possibilities that had not been realised. Transformational creativity changes the space itself, by altering one

of its defining rules, so that ideas that were previously impossible become possible. Boden also distinguished psychological creativity, an idea new to the person who has it, from historical creativity, an idea new to anyone. She argued that computers can exhibit all three kinds, with transformational creativity the hardest to achieve and to evaluate.

L-systems, grammars introduced by Lindenmayer to model the development of organisms, give a concrete conceptual space [6, 7]. A rule such as replacing every segment with a branching pattern generates plant-like forms, as Figure 13.1 shows. Varying the angle or the number of iterations explores the space, while changing the grammar transforms it.

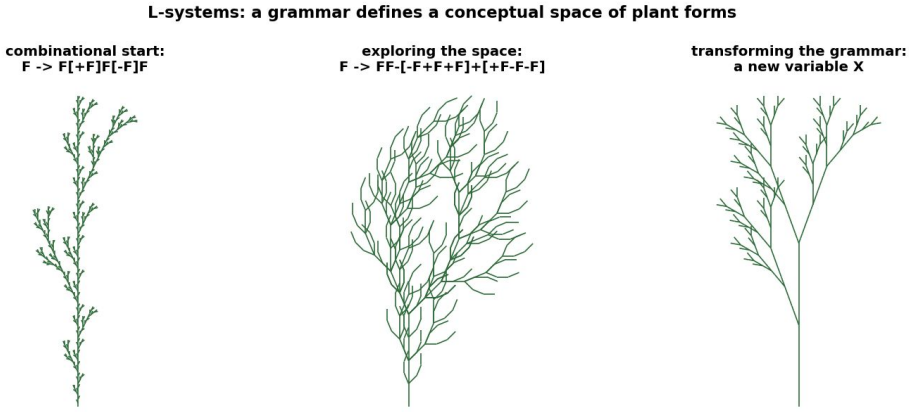


Figure 13.1. Plant forms generated by L-systems. Changing parameters explores the space defined by a grammar, while changing the grammar transforms it [2, 6].

Evaluating creative systems

Ritchie proposed empirical criteria for attributing creativity to a program [3]. The central notions are novelty, quality and typicality. Novelty is the extent to which an output differs from existing examples, and quality is the extent to which it is valued. Typicality is the extent to which it belongs to the intended genre. The criteria are relative to an inspiring set, the examples that guided the construction of the program. Wiggins formalised Boden's account in a creative systems framework that separates the conceptual space, the strategy for traversing it and the evaluation of results [8]. Colton and Wiggins described computational creativity as a field that studies systems that take on responsibilities in creative tasks [9]. They noted that the way a system's process is framed and perceived affects whether audiences regard it as creative.

Searching for novelty

Lehman and Stanley showed that search guided only by the novelty of behaviour, with no objective, can outperform objective-driven search in deceptive problems [10]. In such problems, the path to a good solution leads away from it at first. Novelty search keeps an archive of past behaviours and rewards candidates that are far from them, which makes it a mechanism for exploratory creativity. The interactive lab lets students combine, explore and transform L-system plants and computes novelty against a growing gallery. The Colab notebook compares

novelty search with a search that maximises one objective and measures how much of the space each covers.

> ***Pause and reflect.*** *In the interactive lab, which of your plants would you call creative, and was the judgement yours or the program's? Where does the creativity lie?*

Hands-on components

Interactive lab: Grow plants with an L-system. Explore by changing the angle and the number of iterations or by random variation, combine two gallery rules, or transform the grammar. Novelty is measured against the inspiring set and your gallery, and value is your own rating [2, 3]. Link: https://utkukose.github.io/philosophy-of-ai-NB-lecture/weeks/week-13/lab.html

Colab notebook: It implements L-systems and a turtle interpreter, describes each plant by a few features, and compares novelty search with objective-driven search in the same space of rules [6, 10]. Link: https://colab.research.google.com/github/utkukose/philosophy-of-ai-NB-lecture/blob/main/weeks/week-13/NB13_creativity.ipynb

Reflection prompts

1. Which of your lab plants would you call H-creative, P-creative or not creative at all? Justify your judgements.
2. Does Turing's appeal to surprise answer Lovelace's objection? Consider who is surprised and why.
3. Identify a result in your field that you would call transformational. Could a present AI system have produced it?

Weekly task and submission

What to do. Assess whether one generative AI system is creative. First, choose a system, such as an image generator or a music generator. Second, assess it in each of Boden's three senses: combinational, exploratory and transformational creativity. Third, apply Ritchie's criteria of novelty, quality and typicality to one of its outputs. Support the assessment with one example from the interactive lab or the Colab notebook.

Where to look. Lecture: sections "Three kinds of creativity" and "Evaluating creative systems". Interactive lab: the L-system plants, which you explore, combine and transform (tab: Explore). Colab notebook: Section 3 with Exercise 2, which compares novelty search with objective search.

Example. An image generator that paints a city in the style of a famous painter combines familiar elements, which is combinational creativity. Exploration would search the space of styles more widely, as novelty search does in the Colab notebook. Transformational creativity

would require changing the rules of the space itself, as when you transform the grammar of a plant in the interactive lab.

What to submit. About 500 words, and the Colab notebook with Exercises 1 and 2 completed.

The weekly task supports self-learning and builds a personal portfolio. In an active semester, students who follow the course with the instructor can send their work for evaluation to utkukose@sdu.edu.tr or utkukose@gmail.com. The e-mail contains the items that the task lists under What to submit. It also includes the learning log exported from the interactive lab (tab: Reflect and submit).

References

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